

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a).

$$\begin{bmatrix} 0 & T_{xy} & 0 \\ -T_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xy} = M_{pz}^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} T_{xxx} & 0 & 0 \\ -T_{xxx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -T_{xxx} & 0 \end{bmatrix}$$

$$T_{xxx} = T_{f2}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ T_{yzx} & 0 & 0 \\ 0 & -T_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzx} = 3M_{du}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} T_{yzx} & 0 & 0 \\ 0 & T_{yzx} & 0 \\ 0 & 0 & T_{xyz} \end{bmatrix}$$

$$T_{yzx} = -M_{du}^{(2)} + M_s^{(1)}$$

$$T_{xyz} = 2M_{du}^{(2)} + M_s^{(1)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & -T_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -T_{xxzx} \\ T_{xxzx} & -T_{xxzx} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -T_{xxzx} & 0 & 0 \end{bmatrix}$$

$$T_{xxzx} = T_{ga}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{xxzx} & T_{xxxy} \\ 0 & 0 & 0 & 0 & -T_{xxzx} & -T_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{yzzx} & T_{xxzx} \\ -T_{xxzx} & T_{xxzx} & 0 & -T_{yzzx} & 0 & 0 \\ -T_{xxxy} & T_{xxxy} & 0 & -T_{xxzx} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{xxzx} &= -2M_{f1}^{(1)} \\
T_{xxxy} &= -2M_{faz}^{(1)} + 2M_{pz}^{(1)} \\
T_{yzzx} &= -4M_{faz}^{(1)} - M_{pz}^{(1)}
\end{aligned}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & T_{yzzx} & 0 \\ -T_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzzx} = -M_{pz}^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & T_{xxzx} & T_{xxxy} \\ 0 & -T_{xxzx} & T_{xxxy} \\ 0 & 0 & T_{zzxy} \\ 0 & T_{yzzx} & 0 \\ T_{yzzx} & 0 & 0 \\ T_{xxzx} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{xxzx} &= M_{f1}^{(2)} \\
T_{xxxy} &= -M_{faz}^{(2)} + M_{pz}^{(3)} \\
T_{zzxy} &= 2M_{faz}^{(2)} + M_{pz}^{(3)} + 2M_{pz}^{(4)} \\
T_{yzzx} &= -M_{faz}^{(2)} + M_{pz}^{(4)}
\end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{yzzx} & T_{yzxy} \\ T_{yzxy} & -T_{yzxy} & 0 & T_{yzzx} & 0 & 0 \\ T_{xyxx} & T_{xyxx} & T_{xyzz} & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{yzzx} &= -M_{faz}^{(3)} + M_{pz}^{(6)} \\
T_{yzxy} &= M_{f1}^{(3)} \\
T_{xyxx} &= -M_{faz}^{(3)} + M_{pz}^{(5)} \\
T_{xyzz} &= 2M_{faz}^{(3)} + M_{pz}^{(5)} + 2M_{pz}^{(6)}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{xxzx} & T_{xxxy} & 0 & T_{xxxx} & T_{xxyx} \\ 0 & 0 & 0 & 0 & -T_{xxzx} & -T_{xxyx} & 0 & -T_{xxxx} & -T_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & T_{zzxy} & 0 & 0 & -T_{zzxy} \\ 0 & 0 & 0 & 0 & T_{yzzx} & T_{yzxy} & 0 & T_{yzxz} & T_{yzxy} \\ -T_{yzxy} & T_{yzxy} & 0 & -T_{yzzx} & 0 & 0 & -T_{yzzx} & 0 & 0 \\ -\frac{T_{xxxx}}{2} - \frac{T_{xxyx}}{2} & \frac{T_{xxxx}}{2} + \frac{T_{xxyx}}{2} & 0 & -T_{xxxx} & 0 & 0 & -T_{xxxx} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{xxzx} &= -2M_{f1}^{(1)} + M_{f1}^{(2)} + T_{ga}^{(1)} \\
T_{xxxy} &= -2M_{faz}^{(1)} - M_{faz}^{(2)} + 2M_{pz}^{(1)} + M_{pz}^{(3)} \\
T_{xxzx} &= -2M_{f1}^{(1)} - M_{f1}^{(2)} + T_{ga}^{(1)} \\
T_{xxyx} &= -2M_{faz}^{(1)} + M_{faz}^{(2)} + 2M_{pz}^{(1)} - M_{pz}^{(3)} \\
T_{zzxy} &= 2M_{faz}^{(2)} + M_{pz}^{(3)} + 2M_{pz}^{(4)} \\
T_{yzzx} &= -4M_{faz}^{(1)} - M_{faz}^{(2)} - M_{pz}^{(1)} + M_{pz}^{(4)} \\
T_{yzxy} &= -2M_{f1}^{(1)} - T_{ga}^{(1)} \\
T_{yzxz} &= -4M_{faz}^{(1)} + M_{faz}^{(2)} - M_{pz}^{(1)} - M_{pz}^{(4)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{yzzx} & T_{yzxy} & 0 & T_{yzxz} & T_{yzxy} \\ T_{yzxy} & -T_{yzxy} & 0 & T_{yzxz} & 0 & 0 & T_{yzxz} & 0 & 0 \\ T_{xyxx} & T_{xyxx} & T_{xyzx} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzzx} = -M_{faz}^{(3)} - M_{pz}^{(2)} + M_{pz}^{(6)}$$

$$T_{yzxy} = M_{f1}^{(3)}$$

$$T_{yzzz} = -M_{faz}^{(3)} + M_{pz}^{(2)} + M_{pz}^{(6)}$$

$$T_{xyxx} = -M_{faz}^{(3)} + M_{pz}^{(5)}$$

$$T_{xyzx} = 2M_{faz}^{(3)} + M_{pz}^{(5)} + 2M_{pz}^{(6)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & T_{xxxx} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ -T_{xxxx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ T_{xxxx} & 0 & 0 \\ 0 & 0 & -T_{xxxx} \\ 0 & -T_{xxxx} & 0 \end{bmatrix}$$

$$T_{xxxx} = T_{ga}^{(1)}$$